Journal of Behavioral and Experimental Economics 119 (2025) 102477

Contents lists available at ScienceDirect

Journal of Behavioral and Experimental Economics

journal homepage: www.elsevier.com/locate/jbee

Do incentives matter in elicitation of beliefs?

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ARTICLE INFO

Keywords:
Belief elicitation
Incentives
Bias
Accuracy
Response time

ABSTRACT

Beliefs are a critical component of economic decision-making. Since they are not directly observable, they are generally elicited through survey questions. Hence, it is important to know whether and how incentivization affects the elicited beliefs. Using an experiment, this paper tests the effect of incentivized versus unincentivized belief elicitation on students' beliefs about returns to effort in an educational setting. Initially, incentivization significantly reduces the *bias* in beliefs, without impacting the *accuracy* of beliefs. It also reduces the frequency of round answers and increases time spent responding to the belief questions, both of which suggest increased effort. However, these initial effects on bias and response times diminish or become nonexistent in later weeks, suggesting short-lived effects.

1. Introduction

Beliefs are a critical component of economic decision-making under uncertainty (Manski, 2004), but they are not directly observable. Hence, measuring expectations through surveys, which are often incentivized, has become a widespread practice in economics (Fuster et al., 2022; Stantcheva, 2023), providing insights about individuals' decision-making processes (See Charness et al. (2021), Schlag et al. (2015), Schotter and Trevino (2014) for reviews of belief elicitation in laboratory settings).

A key methodological question is whether incentivized belief elicitation affects the elicited beliefs and their accuracy. Unincentivized belief measures might suffer from a range of issues, including experimenter demand effects, self-serving biases, or a lack of effort in forming and/or reporting beliefs, especially when participants face no cost for inaccurate responses. Incentivized elicitation aims to mitigate these issues by increasing motivation to think carefully about one's beliefs. However, incentivization can also introduce unintended problems. Participants may look up correct answers when incentives are high (Grewenig et al., 2022), misreport strategically to hedge against risk (Armantier & Treich, 2013; Blanco et al., 2010), or display systematic biases, such as reporting beliefs closer to the center of the distribution (Danz et al., 2022). As such, whether and how incentivization affects reported beliefs remains a critical question for experimental design.

This paper investigates the effects of incentivized versus unincentivized belief elicitation about returns to effort in an educational setting. To do so, I recruited college students to learn Spanish for four weeks using an online language learning platform, Duolingo. Students first took an assessment test measuring their baseline knowledge of Spanish. Then, I elicited students' beliefs about returns to Duolingo effort. Specifically, they were asked how much test scores will improve on average if someone completes a certain number of lessons for the four-week period. In a between-subject design, a random subset of subjects answered these questions without any incentives for accuracy (flat-pay condition) whereas the remaining subjects were incentivized based on accuracy using an absolute deviation rule (incentives condition). I elicited students' beliefs multiple times on a weekly basis throughout the study. I also measured how many lessons subjects completed in the four-week period. Students took a final assessment test measuring their Spanish knowledge.[2]

I use an absolute scoring rule, rather than a quadratic scoring rule (Brier, 1950) or a binarized scoring rule (Hossain & Okui, 2013), to reduce the complexity of the instructions and minimize cognitive burden of participants. While the absolute scoring rule is not proper for eliciting the *mean* of a belief distribution — except when the distribution is symmetric and unimodal — it is proper for eliciting the *median* (Jose, 2016) under risk-neutrality. Nevertheless, this paper does not aim to test whether incentivization leads to reported beliefs that

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[1] I appreciate the generous support of the Institute for Research in the Social Sciences at Stanford University, Leonard W. Ely and Shirley R. Ely Graduate Student Fellowship, Russell Sage Foundation, Shultz Graduate Student Fellowship in Economic Policy, and Stanford Graduate Research Opportunities Fund. This research is approved by Stanford Panel on Non-Medical Human Subjects (IRB protocol 36512).

[2] After the initial belief elicitation, students were randomly assigned to one of the treatments that aimed to manipulate their beliefs about returns to Duolingo effort. Ersoy (2023) investigates the effect of these treatments on beliefs (pooling incentivized and unincentivized beliefs), Duolingo effort, and test performance.

https://doi.org/10.1016/j.socec.2025.102477
Received 10 April 2025; Received in revised form 31 October 2025; Accepted 1 November 2025
Available online 8 November 2025
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more closely reflect participants' true beliefs. Rather, the focus is on whether incentives affect the alignment between reported beliefs and the corresponding benchmarks.

This paper investigates whether incentivizing beliefs leads to beliefs that are closer to the benchmark: the causal effect of effort on performance, estimated using instrumental variables method with experimental data from Ersoy (2021), in which participants were randomly assigned different numbers of lessons. I hypothesize that incentives will result in *less* biased beliefs — measured as the difference between the reported belief and the benchmark — and *more* accurate beliefs, measured as the absolute difference between the two. The underlying mechanism is that incentives increase cognitive effort in belief formation and reporting. I also hypothesize that incentivized participants will report greater confidence in their beliefs, either because they exert more effort in forming or reporting them or because their beliefs are in fact more accurate.

I find that while students overestimated the returns to Duolingo effort in both flat-pay and incentives conditions, the bias (belief-benchmark) was significantly lower at the baseline when the subjects were incentivized. The initial belief distributions were also significantly different from each other. The difference in belief distributions across the two conditions persisted in the first follow-up survey (week 1), but there were no significant differences in bias or belief distributions at the later surveys (weeks 2 through 4). Incentivized students spent more time on belief elicitation screen at the baseline, suggesting greater cognitive engagement, but this effect also disappeared in subsequent weeks. Additionally, subjects in the incentives condition have fewer round answers to the belief questions in all weekly surveys, which again suggest more effort in answering these questions. Despite the reduction in bias and increase in effort at the baseline, incentivization did not significantly enhance the overall accuracy of beliefs (absolute value of bias) at the baseline. It also did not influence students' confidence in the accuracy of their beliefs. The findings suggest that while incentives lead to less biased beliefs on average, they might not be enough to improve accuracy of beliefs. Furthermore, the effects of incentives can be short-lived.

This paper contributes to belief elicitation methods literature by exploring the importance of incentivization on the stated beliefs. While the impact of incentivizing beliefs has been well-explored in laboratory-based economic games (Gächter & Renner, 2010; Wang, 2011), there is a smaller literature trying to understand how incentives influence survey-based measures of beliefs about events. Using subjects from MTurk, Roth and Wohlfart (2020) studies whether measuring beliefs about likelihood of a recession using an incentivized belief elicitation (quadratic scoring rule) affects reported beliefs and finds that the distributions of incentivized and unincentivized beliefs are not statistically distinguishable. Stantcheva (2021) shows providing monetary incentives does not affect misperceptions of US survey respondents. In a representative online survey in Germany, Grewenig et al. (2022) tests the effect of incentivizing belief accuracy using a simple method (rewarding the subjects with a bonus if their guess is roughly correct) on beliefs about average earnings by professional degree and average public school spending and finds that incentivization does not impact beliefs regarding earnings, but improves school-spending related beliefs. In their review paper, Haaland et al. (2023) concludes that incentives do not affect accuracy of belief elicitation in non-political domains or when the answers are not easily accessible. This paper contributes to this literature by showing that offering sizeable incentives for belief accuracy can lead to decreases in biases in beliefs regarding returns to effort without necessarily impacting their accuracy and that these effects can be short-lived.

2. Experimental design

Subjects were asked to provide their beliefs about returns to effort in Duolingo for the participants of a previous experiment (Ersoy, 2021). In particular, they were asked the following question: "On average, how much improvement in test scores do you think results from completing X lessons per week for 4 weeks (X*4 lessons in total)?" for five different levels of effort (8, 12, 16, 20, and 24 lessons). Subjects answered this question five different times during the study (baseline, week 1, week 2, week 3, week 4). Each week, I also elicited participants' confidence in the accuracy of their beliefs using a 5-item Likert scale question (1 being extremely accurate and 5 being not accurate at all).

Participants do not receive any direct feedback about the accuracy of their beliefs until the end of the experiment except the information they receive if they were in one of the information treatments. In the Control treatment (No Information), students do not receive any information about the effort-performance relationship. In three treatments (Average Information), students learn about the *average* causal relationship for the previous participants. These treatments provide information about (1) improvement in test scores after completing 8 lessons per week for 4 weeks (partial information-8 lessons treatment), (2) improvement in test scores after completing 24 lessons per week for 4 weeks (partial information-24 lessons treatment), and (3) improvement in test scores after completing 8, 16, and 24 lessons per week for 4 weeks (full information treatment). Ersoy (2021) manipulated Duolingo effort by assigning participants different numbers of lessons to measure the causal returns to effort via instrumental variables estimation. These estimated returns were then used to construct the information provided in these treatments as well as to calculate the benchmark used in this paper. In the Low (High) treatment, students receive anecdotal information based on a single student from the previous experiment whose returns were unusually low (high) and whose pre-test score was similar to the score of the recipient of the information. The information was provided after the baseline belief elicitation and students did not know they would receive this information beforehand. Hence, subjects in all treatments received the exact same instructions in the baseline survey.

Before students provide their beliefs, students were provided with the details about the Duolingo lessons and the initial test score of the average participant (289 out of 1000) in the previous experiment. Students were given the interpretation of test scores (for example, a person who has perfectly mastered Duolingo Beginner skills should be able get a score of 338 points, etc.). Students were able to access these details on the belief elicitation screen at each weekly survey.[3] Furthermore, students took the baseline Spanish test before belief elicitation; hence, they were familiar with the test.

A subset of the subjects were randomly assigned to receive bonus payment based on accuracy of their answers (incentives condition) and the others did not receive any bonus payment (flat-pay condition). Beliefs were elicited in the same way under the incentives condition and flat-pay condition. In the instructions of the belief survey, subjects who were assigned to the incentives condition were told that the accuracy of their answers to the belief questions will affect the amount of bonus payment they will get from the study. To maximize salience of the incentives, students were reminded of the incentives on the belief-elicitation screen and were able to access more details about the bonus payment.[4] Appendix Fig. A.1 provides the screenshot of the belief questions highlighting the differences between the two conditions.

For those who were in the incentives condition, one of the five belief questions in one of the five weekly surveys was randomly chosen at the end of the study for bonus payment calculation. Baseline survey was

[3] During the baseline survey, 24.7% of students clicked the button on the belief elicitation screen to learn more about what test scores mean, while 16.9% clicked to view an overview of Duolingo lessons.

[4] During the baseline survey, 10.8% of students in the incentives condition clicked the button to learn more about the bonus payment on the belief-elicitation screen, while this percentage ranged from 1% to 4% in the later surveys.

chosen with 80% chance and each of the other surveys had 5% chance of being selected.[5] Subjects in the incentives condition were fully informed about this procedure. Appendix Fig. A.2 provides the screenshot of these instructions. For each point subjects' beliefs differ from the benchmark for the chosen belief question, $0.50 was subtracted from the bonus payment of $30 until the bonus payment reaches $0 (an absolute scoring rule).

Under the flat-pay condition, subjects were paid $50 if they completed all the surveys and tests throughout the experiment. Under the incentives condition, subjects received $40 as long as they completed all the surveys and tests plus a bonus payment of up to $30.[6] The payment amounts were determined to make the expected value of payments to be equal in the flat-pay and incentives conditions. Average final payment was $42.56 (standard deviation is $15) in the flat-pay condition and $40.06 (standard deviation is $12.52) in the incentives condition. I am unable to reject the equality of the means between the two conditions using a two-sided t-test (p-value is 0.259).[7] All payments were paid as Amazon e-gift cards at the end of the study.

3. Summary statistics and balance

Appendix Table A.1 Column 1 presents the mean values of the 52 control variables for the participants of the study. 76% of the participants were female, 42% was white and 31% was Asian. Average age was 24 and average college GPA was 3.38. 52% was enrolled in Health and STEM fields and 31% was enrolled in Humanities and Social Sciences. 92% of them reported not knowing any Spanish or knowing Spanish only at the beginner level. Average family income was around 81K. Columns 2 and 3 present the mean values of the control variables for students in the flat-pay and incentives conditions, respectively. Column 4 tests for balance between the two conditions and reports the p-values from the two-sided t-test for equality of the means. Null hypothesis of equality across conditions is rejected for 4 out of 52 variables at the 5% level (the flat-pay group was more likely to be Asian, less likely to be undecided about their major, less conscientious, and had less self-control than the incentives group) and for another 4 variables at the 10% level (the flat-pay group was less likely to be in the "other" race category, had lower GPA and higher family income, and was more likely to be composed of seniors than the incentives group). One can also reject the equality of the distribution of the p-values to a uniform distribution, which raises concerns about balance. Hence, I present the results both without any controls and with an extensive set of controls in the next section. Crucially, as Appendix Table A.1 shows survey non-response was similar across the two conditions.

4. Findings

In this section, I first present the results for bias in baseline beliefs for each of the five different levels of effort and for a composite measure of beliefs. After discussing bias in baseline beliefs, I investigate the bias in beliefs during the follow-up period. Then, I present the results regarding accuracy of beliefs. Next, I explore whether incentivizing beliefs make subjects put more effort in forming and/or reporting beliefs, using response times and frequency of round answers as proxies. Finally, I investigate whether incentivization affects confidence in beliefs.

First, I run the following ordinary least squares (OLS) regression to investigate how incentivization affects bias:

$$\hat{b}_{ij,baseline} - \theta_j = \alpha_0 + \alpha_1 Incentives_i + \Gamma \mathbf{X_i} + \epsilon_{ij,baseline} \text{ for } j \in 8, 12, 16, 20, 24 \quad (1)$$

where $\hat{b}_{ij,baseline}$ is the baseline reported belief of individual i regarding improvement in test scores for effort level j (8, 12, 16, 20, and 24 lessons per week for four weeks). Since the highest possible belief one should hold for the improvement in test scores for any effort level is 711 points (maximum possible test score (1000 points) minus the average initial score of previous participants (289 points)), I top coded the beliefs that are higher than 711 points as 711 points.[8] θ_j is the estimated benchmark, which is calculated via instrumental variables estimation using data from Ersoy (2021).[9] I define bias as the difference between the reported belief and the estimated benchmark. $Incentives_i$ is a dummy variable that is equal to 1 if the subject i is randomly assigned to receive payments based on the accuracy of their beliefs and 0 otherwise. $\mathbf{X_i}$ is a vector of individual characteristics.

Table 1 Panel A presents the results of Eq. (1) with no additional controls. Baseline beliefs under the incentives condition are less biased than the baseline beliefs under the flat-pay condition. We can reject the equality of the means between the flat-pay and incentives conditions at the 10% level for all effort levels, except for 8 lessons. Panels B and C show that the results continue to hold with the addition of the basic and extensive controls, though they lose significance for lower levels of effort.

Appendix Fig. A.3 shows the cumulative distribution of beliefs separately under unincentivized and incentivized belief elicitation for five different levels of effort. P-values for the two-sample Kolmogorov–Smirnov test for equality of distribution functions is 0.016 for 8 lessons, 0.005 for 12 lessons, 0.031 for 16 lessons, 0.051 for 20 lessons, and 0.124 for 24 lessons, suggesting that these distributions are different.

To study beliefs concisely, I created a per-lesson belief measure which averages the students' beliefs about improvement in test scores for five different levels of effort (8, 12, 16, 20, and 24 lessons per week for four weeks) divided by the corresponding effort level. Then, I estimate the following OLS regression:

$$\bar{\hat{b}}_{i,baseline} - \bar{\theta} = \alpha_0 + \alpha_1 Incentives_i + \Gamma \mathbf{X_i} + \epsilon_{i,baseline} \quad (2)$$

where $\bar{\hat{b}}_{i,baseline}$ is the per-lesson reported belief of individual i at the baseline and $\bar{\theta}$ is the estimated per-lesson benchmark (2.46 points).

As Table 2 Column 1 Panel A shows, mean bias at the baseline is 2.6 points under the flat-pay condition (p-value is less than 0.001), meaning that individuals in the flat-pay condition, on average, overestimate the returns to Duolingo effort by 2.6 points compared to the estimated benchmark. Mean bias is 0.99 points lower under the incentives condition compared to the flat-pay condition (p-value is 0.048). Table 1 Column 1 Panels B and C show that this result continues to hold with the addition of the basic and extensive controls.

[5] This design choice was mainly for eliminating subjects' paying attention to the information treatments for spurious reasons. Effect of information treatments on beliefs was the main focus of Ersoy (2023).

[6] Subjects who did not complete the final Spanish test received only half of their fixed payment: $25 in the flat-pay condition and $20 in the incentives condition. In addition, $5 was deducted from the fixed payment for each weekly belief survey that was missed. Subjects in the incentives condition could still earn their full bonus payment even if they missed some weekly surveys (as long as the survey selected for payment was completed) or did not complete the final Spanish test.

[7] For subjects who completed the final Spanish test, average payment was $49.70 (standard deviation is $1.21) in the flat-pay condition and $45.04 (standard deviation is $9.01) in the incentives condition. We can reject the equality of the means between the two conditions using a two-sided t-test (p-value is 0.0035) for this group.

[8] For baseline beliefs, none of the 8-lesson or 12-lesson beliefs were above 711 points. Less than 1% of 16-lesson beliefs, less than 5% of the 20-lesson beliefs, and less than 12% of the 24-lesson beliefs were above 711 points. Conducting the analysis with the original data does not change the results quantitatively or qualitatively.

[9] Using the *median* improvement in test scores (estimated via quartile instrumental variables method using data from Ersoy (2021)) does not change the results.

Table 1
Effects of incentives on bias in baseline beliefs.

	(1) 8-lesson	(2) 12-lesson	(3) 16-lesson	(4) 20-lesson	(5) 24-lesson
Panel A: No Controls					
Incentives	−37.21	−60.85**	−59.28*	−68.38**	−71.73*
	(23.60)	(28.41)	(32.43)	(34.53)	(37.06)
Constant	85.84***	133.6***	164.2***	195.8***	227.0***
	(21.62)	(26.00)	(29.45)	(31.23)	(33.29)
Panel B: Demographic and Education Controls					
Incentives	−36.66	−68.14**	−70.93**	−87.61**	−99.33***
	(23.31)	(28.64)	(32.41)	(34.28)	(36.77)
Constant	229.2*	22.77	−64.38	−52.50	−80.87
	(134.4)	(157.2)	(172.2)	(178.2)	(192.8)
Panel C: All Controls					
Incentives	−25.40	−49.13	−47.63	−71.36**	−84.48**
	(24.78)	(30.29)	(33.91)	(35.52)	(37.81)
Constant	−1679.5	−1720.8	−2124.7*	−2200.0*	−2557.8*
	(1140.9)	(1230.8)	(1229.3)	(1261.1)	(1365.5)
Observations	243	243	243	243	243

Dependent variable is the bias in beliefs (difference between belief about improvement in test scores and the benchmark) elicited in the baseline survey. Columns 1, 2, 3, 4, and 5 belong to 8-lesson, 12-lesson, 16-lesson, 20-lesson, and 24-lesson belief measures, respectively. Panel A does not include any controls. Panel B includes age, gender, race category, college GPA, ACT/SAT math and verbal scores, whether ACT/SAT scores are missing, respondents' mother's and father's years of education, and student status (freshman, sophomore, junior, senior, master's, phd, other). Panel C includes big 5 personality traits, measures of growth mindset, self control, and locus of control, measures of risk aversion, present biasedness, and patience, annual income and monthly expenditure, commitment to learning Spanish and interest in learning Spanish, whether the respondent is learning Spanish for business purposes, whether the respondent is native English speaker, and all controls included in Panel B. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2
Effects of incentives on bias in per-lesson beliefs.

	(1) Baseline	(2) Week 1	(3) Week 2	(4) Week 3	(5) Week 4
Panel A: No Controls					
Incentives	−0.992**	−0.835*	−0.309	−0.211	−0.0504
	(0.500)	(0.480)	(0.414)	(0.414)	(0.376)
Constant	2.569***	2.396***	1.603***	1.156**	1.111**
	(0.454)	(0.621)	(0.552)	(0.539)	(0.506)
Panel B: Demographic and Education Controls					
Incentives	−1.161**	−0.558	−0.0914	0.0186	0.214
	(0.500)	(0.500)	(0.439)	(0.419)	(0.420)
Constant	1.026	3.064	3.543**	1.917	3.112
	(2.648)	(2.891)	(1.743)	(2.269)	(2.028)
Panel C: All Controls					
Incentives	−0.867*	−0.481	0.0652	0.314	0.446
	(0.521)	(0.529)	(0.476)	(0.481)	(0.460)
Constant	−35.14*	−6.649	−0.891	−7.714	1.867
	(21.13)	(18.69)	(15.92)	(18.01)	(17.11)
Observations	243	232	216	212	199

Dependent variable is the difference between per-lesson belief about improvement in test scores and the per-lesson benchmark. Different columns refer to the belief elicitation at different weeks. All columns, except column 1 include information treatment dummies. Panel A does not include any additional controls. Panel B includes demographic and education controls listed in Table 1. Panel C includes all controls listed in Table 1. The number of observations across the columns differ because not all participants took all the weekly belief surveys. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Fig. 1 shows the cumulative distribution of beliefs, separately under unincentivized and incentivized belief elicitation. 80% of the subjects in the flat-pay condition had a baseline per-lesson belief that was above the estimated per-lesson benchmark whereas 60% of the subjects in the incentives condition had a baseline per-lesson belief that was above the per-lesson benchmark. Conversely, 20% of the subjects in the flat-pay condition had a baseline per-lesson belief that was below the estimated per-lesson benchmark whereas 40% of the subjects in the incentives condition had a baseline per-lesson belief that was below the per-lesson benchmark. The P-value for the two-sample Kolmogorov–Smirnov test for equality of distribution functions is equal to 0.016, leading the rejection of the equality of the distributions.[10]

Given the difference in the baseline beliefs under the two conditions, a natural follow-up question is whether these differences persist beyond baseline. To investigate this question, I look at the difference between the reported per-lesson beliefs and the per-lesson benchmark in Weeks 1 through 4 using the following OLS regression:

$$\bar{\tilde{b}}_{iw} - \bar{\theta} = \alpha_0 + \alpha_1 Incentives_i + \sum_{k=2}^{6} \beta_{ik} Treatment_{ik} + \Gamma \mathbf{X_i} + \epsilon_{iw} \text{ for } w \in 1,2,3,4 \tag{3}$$

where $\bar{\tilde{b}}_{i,w}$ is the per-lesson reported belief of individual i in week w survey and $\bar{\theta}$ is the estimated per-lesson benchmark (2.46 points). $Treatment_{ik}$ are the dummy variables for each information treatment and the no information treatment is the omitted category. In this analysis, I control for the information treatment dummies since students

[10] I am unable to reject that incentives have no effect on the dispersion of beliefs. The standard deviation of average baseline belief is 2.90 in the flat-pay condition which is not statistically significantly different from the respective value of 2.95 in the incentives condition.

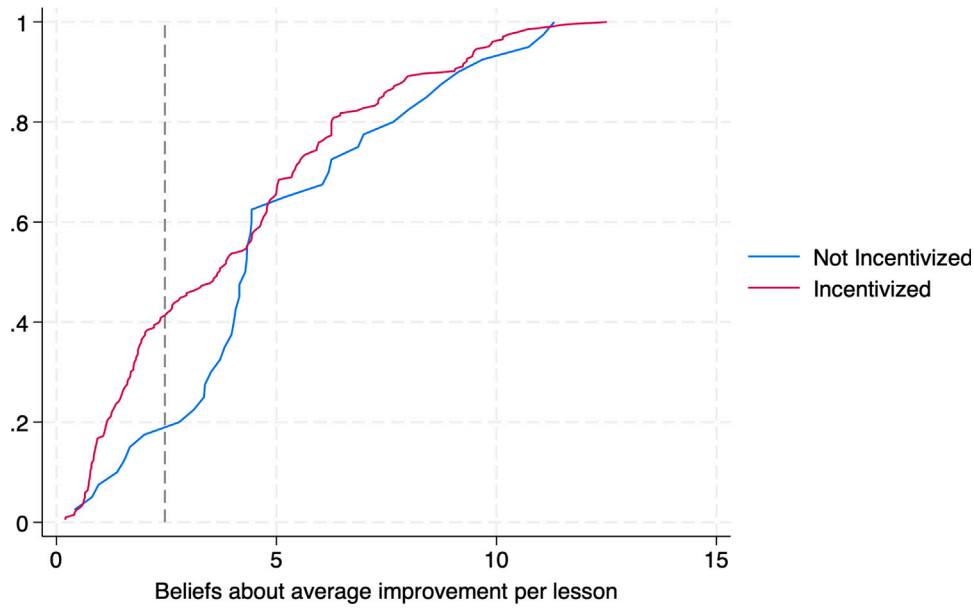


Fig. 1. Distribution of baseline beliefs with and without incentivization.
Notes: Gray dashed line is the accuracy benchmark estimated using data from Ersoy (2021) via instrumental variables estimation. 80% of the subjects in the flat-pay condition had a baseline belief that was above the benchmark whereas 60% of the subjects in the incentives condition had a baseline belief that was above the benchmark.

were subject to different information treatments after baseline belief elicitation.

Table 2 Columns 2-5 present the results. In Week 1 belief elicitation, students who are in the incentives condition are still less biased compared to students in the flat-pay condition as Panel A shows (p-value is 0.083). But this difference in means loses its significance once I include the basic or extensive set of controls. The P-value for the two-sample Kolmogorov–Smirnov test for equality of distributions for the per-lesson belief measure (Appendix Fig. A.4) is equal to 0.046. We can also reject the equality of distributions for all effort levels (Appendix Fig. A.5), except for beliefs about 8 lessons. For the other weeks (weeks 2 through 4), the means and the distributions are not different under the two conditions.[11]

One potential reason for the disappearing differences between the incentives and flat-pay conditions in the later weeks is the fact that information was provided after baseline belief elicitation in information treatments. Ersoy (2023) shows that information treatments shift students' beliefs about returns to effort toward the information they receive. As a result, subjects in the Average Information treatments are likely to report beliefs closer to the corresponding benchmarks, while those in the Low or High treatments receive information that moves their beliefs away from the benchmark. Subjects in the No Information treatment, by contrast, receive no such guidance. A cleaner approach would be to restrict the analysis to individuals who never received any information (i.e., those in the No Information treatment). However, the small sample size in this group prevents a meaningful analysis.

I hypothesized that incentives will lead to more accurate beliefs, defined as the absolute difference between the stated beliefs and the benchmark. Appendix Table A.2 displays the results regarding this measure using OLS regressions similar to Eqs. (2) and (3) where the dependent variable is the absolute value of the bias instead of the bias. Beliefs are inaccurate in the flat-pay condition and incentivizing beliefs does not statistically affect the accuracy of the beliefs either at the baseline or during the following weeks. The results remain similar if I include a basic or extensive set of controls. I am also unable to reject the equality of the distributions of accuracy of beliefs under the incentives and flat-pay conditions in any week.

I hypothesized that incentivization will lead less biased and more accurate beliefs by increasing the effort subjects put into forming and/or reporting their beliefs. A potential proxy for effort is the time spent on the belief elicitation screen. To study the effects of incentivization on response times, I run the following OLS regressions:

$$responseTime_{i,baseline} = \alpha_0 + \alpha_1 Incentives_i + \Gamma \mathbf{X_i} + \epsilon_{i,baseline} \tag{4}$$

$$responseTime_{i,w} = \alpha_0 + \alpha_1 Incentives_i + \sum_{k=2}^{6} \beta_{ik} Treatment_{ik} + \Gamma \mathbf{X_i} + \epsilon_{iw} \quad \text{for } w \in 1,2,3,4 \tag{5}$$

Table 3 displays the results week by week. As Panel A Column 1 shows, the average response time was 140 s under the flat-pay condition. The average response time under the incentives condition was almost 3 min longer than the flat-pay condition (p-value is less than 0.001) at the baseline belief elicitation, suggesting that incentives make students think more carefully about the belief questions.[12] This result remains significant at the 1% level after controlling for basic covariates (Panel B) and an extensive set of covariates (Panel C). Since average response times can be sensitive to the outliers, I also look at the median response times. Median response time is 1 min and 47 s for the flat-pay condition and 3 min for the incentives condition. Equality of medians is rejected via Chi-Squared Test (p-value is 0.017). We also find that fast responses (defined as responses at or below the 25th percentile of the

[11] Since participants in some information treatments were provided with information regarding average returns to Duolingo effort after the baseline survey, we might expect the effects of incentivization on bias to be different for this subsample. Hence, we repeat the analysis restricting the sample to the subset of participants who received information about average returns to Duolingo effort (partial information-8 lessons, partial information-24 lessons, and full information). We find that the results are qualitatively similar with effects observed in Week 1, but not in the other weeks.

[12] Since response times are subject to the outlier issues, I dropped the observations that are above the 99th percentile in each week.

Table 3
Effects of incentives on response times.

	(1) Baseline	(2) Week 1	(3) Week 2	(4) Week 3	(5) Week 4
Panel A: No Controls					
Incentives	179.7***	16.18	−0.447	12.54	16.86
	(30.52)	(29.51)	(18.11)	(25.92)	(32.77)
Constant	140.3***	93.26***	82.41***	62.17**	27.95
	(16.08)	(27.65)	(20.89)	(24.34)	(27.76)
Panel B: Demographic and Education Controls					
Incentives	165.8***	14.25	−4.052	14.69	19.38
	(32.94)	(32.17)	(17.65)	(27.84)	(34.34)
Constant	−382.3	−22.46	−134.8	−55.61	376.0*
	(251.5)	(166.7)	(129.4)	(92.41)	(207.3)
Panel C: All Controls					
Incentives	147.0***	15.15	−1.862	22.61	48.38
	(41.89)	(35.99)	(18.38)	(24.17)	(53.60)
Constant	−1589.8	1730.3	−9.372	363.6	−7069.2
	(4093.4)	(1677.6)	(817.6)	(996.4)	(6201.1)
Observations	241	230	214	210	198

Dependent variable is the time spent on the belief elicitation screen. Different columns refer to belief elicitation at different weeks. Since response times are subject to outlier issues, I dropped the observations that are above the 99th percentile in each column. All panels & columns except column 1 include information treatment dummies. Panel A does not include any additional controls. Panel B includes demographic and education controls listed in Table 1. Panel C includes all controls listed in Table 1. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

response time distribution) are less frequent in the incentives condition compared to the flat-pay condition at the baseline.

Appendix Fig. A.6 shows the cumulative distribution of response times across flat-pay and incentives conditions at the baseline belief elicitation. The P-value for the two-sample Kolmogorov–Smirnov test for equality of distribution functions is equal to 0.002, also leading us to reject the equality of the distributions of response times across the flat-pay and incentives conditions.

We do not observe any differences in mean response times across the two conditions for Weeks 1 through 4 (Table 3 Columns 2–5). We also do not observe any differences in distributions or medians across the two conditions for Weeks 1, 2, and 4, and we observe only marginal differences for Week 3.

If response times are a good proxy for effort and if effort reduces bias in beliefs, we would expect a negative correlation between response time and bias. Appendix Fig. A.7 shows this relationship at baseline. Although the correlation coefficient is negative (−0.032), it is small and not statistically significant (p-value is 0.620), which sheds doubt on this evidence. However, when focusing on fast responses (defined as those at or below the 25th percentile of the response time distribution), we find a positive and statistically significant relationship: baseline average beliefs are 0.92 points more upwardly biased with no controls and 1.21 points more biased with the full set of controls (both significant at the 5% level). This suggests that fast responses may be associated with less careful belief formation or reporting, leading to more biased beliefs.

Although there is a clear difference in response times under the flat-pay and incentives conditions in the baseline survey, this does not necessarily reflect a higher level of effort in forming and/or reporting beliefs. An alternative explanation is that subjects in the incentives condition spent more time trying to understand how the incentives will be calculated and that was why they had longer response times in the baseline belief survey. To minimize this possibility, I provide the subjects with the instructions and explanation before the belief elicitation screen. Furthermore, we observe that only 10.8% of students in the incentives condition clicked the button to learn more about the bonus payment on the belief-elicitation screen. However, the lack of correlation between the response times and bias and disappearance of the difference in response times across conditions in the later weeks indicate that this alternative explanation might be a reason for the differences in response times across conditions in the baseline survey.

Another potential proxy for effort is the frequency of the round answers to the belief questions. To study the effects of incentivization on frequency of round answers, I run the following OLS regressions:

$$roundAnswer_{i,baseline} = \alpha_0 + \alpha_1 Incentives_i + \Gamma \mathbf{X_i} + \epsilon_{i,baseline} \tag{6}$$

$$roundAnswer_{i,w} = \alpha_0 + \alpha_1 Incentives_i + \sum_{k=2}^{6} \beta_{ik} Treatment_{ik} + \Gamma \mathbf{X_i} + \epsilon_{iw} \quad \text{for } w \in 1,2,3,4 \tag{7}$$

where $roundAnswer_i$ is equal to 1 if the subject i reported a belief that is multiple of fifty and 0 otherwise. We have 5 observations for each subject at each weekly survey since they were asked the belief questions about 5 different levels of effort (8, 12, 16, 20, and 24 lessons). Standard errors are clustered at the subject level.

Table 4 presents the results. Panel A Column 1 shows that 60% of the belief questions at baseline had round answers in the flat-pay condition. In comparison, round answers were 18.9 percentage points less common in the incentives condition. This pattern remains when additional covariates are included and is also observed in the later weeks.[13] There is a strong positive correlation between bias in a given belief question and whether the answer to that belief question is round (defined as a multiple of 50), and these correlations are statistically significant at the 1% level in all weeks, regardless of whether the analysis includes controls or not. Although suggestive, these findings imply that subjects in the incentives condition exerted greater effort in forming or reporting their beliefs, leading to less biased beliefs.

Finally, I hypothesized that incentivization might increase individuals' confidence in their beliefs. Appendix Table A.3 investigates whether subjects' confidence about the accuracy of their beliefs differs across the two conditions. Dependent variable is equal to 1 if the student chose "extremely accurate", "very accurate", or "moderately accurate" as answer to the confidence question and is equal to 0 if "slightly accurate" or "not accurate at all" was chosen by the student. According to this measure, 58% to 67% of the students are confident about their belief accuracy on a given week. Looking at the raw data, there are no statistically significant differences in confidence across the flat-pay and incentives conditions in any given week (Panel A). Including basic

[13] The results are robust to using multiples of 25 or 100.

Table 4
Effects of incentives on round answers to belief questions.

	(1) Baseline	(2) Week 1	(3) Week 2	(4) Week 3	(5) Week 4
Panel A: No Controls					
Incentives	−0.189***	−0.173***	−0.120**	−0.166***	−0.150***
	(0.0625)	(0.0563)	(0.0534)	(0.0549)	(0.0548)
Constant	0.605***	0.607***	0.502***	0.606***	0.580***
	(0.0574)	(0.0813)	(0.0731)	(0.0665)	(0.0676)
Panel B: Demographic and Education Controls					
Incentives	−0.238***	−0.183***	−0.0900*	−0.170***	−0.145***
	(0.0632)	(0.0568)	(0.0513)	(0.0517)	(0.0554)
Constant	0.748**	0.585	0.379	0.353	0.829**
	(0.353)	(0.366)	(0.385)	(0.346)	(0.329)
Panel C: All Controls					
Incentives	−0.226***	−0.216***	−0.0930*	−0.149***	−0.103**
	(0.0621)	(0.0578)	(0.0533)	(0.0515)	(0.0488)
Constant	−0.415	2.623	−2.549	−2.726	−3.387*
	(2.008)	(2.015)	(2.148)	(1.897)	(1.781)
Observations	1215	1160	1080	1060	995
Subjects	243	232	216	212	199

Dependent variable is the whether a subject reported a belief that is multiple of 50. We have 5 observations for each subject per week (beliefs for 5 effort levels). Different columns refer to the belief elicitation at different weeks. All columns, except column 1 include information treatment dummies. Panel A does not include any additional controls. Panel B includes demographic and education controls listed in Table 1. Panel C includes all controls listed in Table 1. The number of subjects across the columns differ because not all participants took all the weekly belief surveys. Standard errors are clustered at the subject level and are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

controls (Panel B) or extensive controls (Panel C) leads to statistically significant differences only in Week 1 across the two conditions. In particular, subjects under the incentives condition are less confident about their beliefs compared to subjects in the flat-pay condition. But given the non-existent differences in the baseline belief elicitation, this result should be interpreted with caution.

5. Conclusion

Beliefs are key components of economic decisions, but their unobservable nature makes researchers often resort to surveys to measure beliefs. In this paper, I examine how incentivizing accuracy of beliefs impact subjects' beliefs about returns to Duolingo effort. Subjects' beliefs were elicited five times in a 4-week study during which subjects learn Spanish with Duolingo. One group of subjects was randomly assigned to receive incentives for accurate beliefs (the more accurate the belief, the higher the bonus) whereas the other group received a flat-pay. I find that both groups overestimated the returns to effort at the baseline survey, but the bias was significantly smaller in the incentives group, which also spent significantly more time answering the belief questions and gave fewer round answers. However, this reduction in bias did not persist beyond the first follow-up, and incentivization did not lead to significant improvements in belief accuracy or confidence.

My findings are mostly in line with the literature. Similar to the Grewenig et al. (2022), I find that incentives increase the time spent answering belief questions, suggesting increased attention or cognitive effort.[14] Similar to previous studies reviewed in Stantcheva (2023), I also find that incentivization does not affect the accuracy of beliefs when they are first elicited or during the follow-up elicitations. Incentivization also does not significantly impact confidence in the accuracy of beliefs similar to Grewenig et al. (2022), despite incentives being much higher in my setting.[15]

In my setting, incentivization reduced the biases in beliefs, but these effects were short-lived. One potential reason for short-lived effects is that students' familiarity with Duolingo lessons and the context increased during the study period. Additionally, students in three of the information treatments received information about the average returns to Duolingo effort, which might further contribute to the disappearance of the differences in bias across the two conditions. Another potential reason is the way I incentivized the beliefs: The baseline survey was chosen with 80% chance and each of the other surveys had 5% chance of being selected for payment. This design choice might have caused the disappearance of the differences in the later weeks.

Future research could explore whether subjects' familiarity with the context of the elicited beliefs or the existence of motivated beliefs (Zimmermann, 2020) impact the effect of incentivization on elicited beliefs.

CRediT authorship contribution statement

Fulya Y. Ersoy: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Chat GPT in order to revise the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Appendix

A.1. Tables

See Tables A.1–A.3.

A.2. Figures

See Figs. A.1–A.7.

Data availability

Data will be made available on request.

[14] An alternative interpretation is that increase in time is due to time spent understanding the payment rule under the incentives treatment rather than more time thinking about the answer.

[15] Incentives for accurate beliefs were about 0.30 euros in Grewenig et al. (2022).

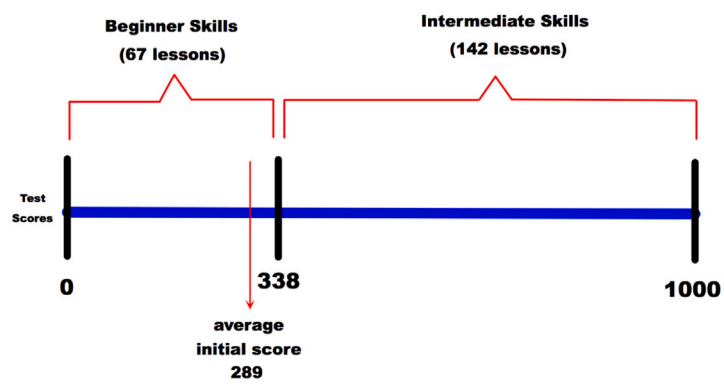
Flat-Pay Condition

Now, we will ask your guess about the improvement of the participants of the previous round.

Incentives Condition

Now, we will ask your guess about the improvement of the participants of the previous round.

The better your guess for the following questions, the higher your bonus payment will be. Please answer the questions as well as you can.



On average, how much improvement in test sores do you think results from completing **8** lessons per week for 4 weeks (**32** lessons in total)?

By improvement, we mean the difference between the final test score and the initial test score. (Please only type in numbers, no words or blanks!)

On average, how much improvement in test sores do you think results from completing **12** lessons per week for 4 weeks (**48** lessons in total)?

By improvement, we mean the difference between the final test score and the initial test score. (Please only type in numbers, no words or blanks!)

Fig. A.1. Belief questions.

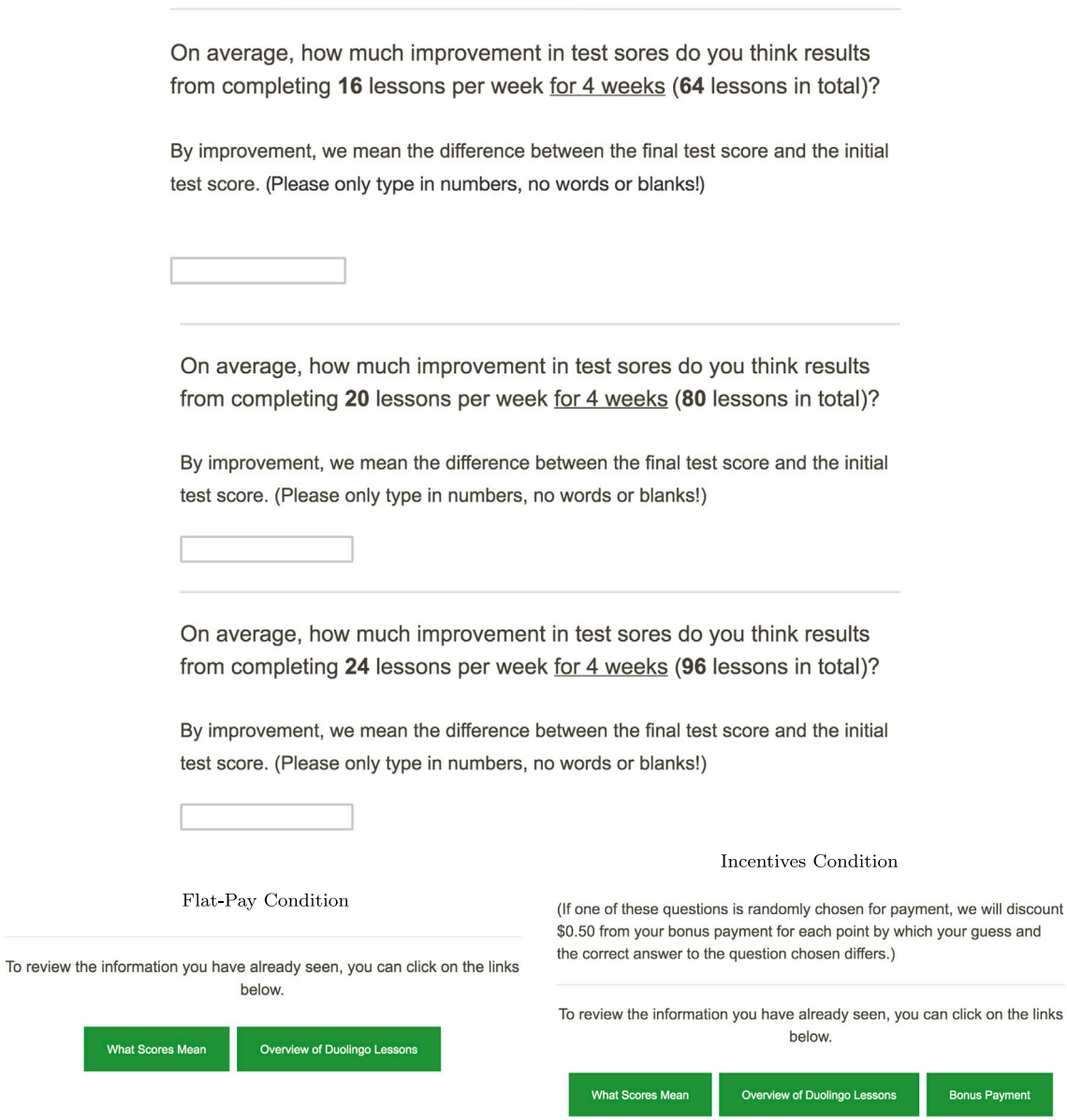


Fig. A.1. (*continued*).

Your Task

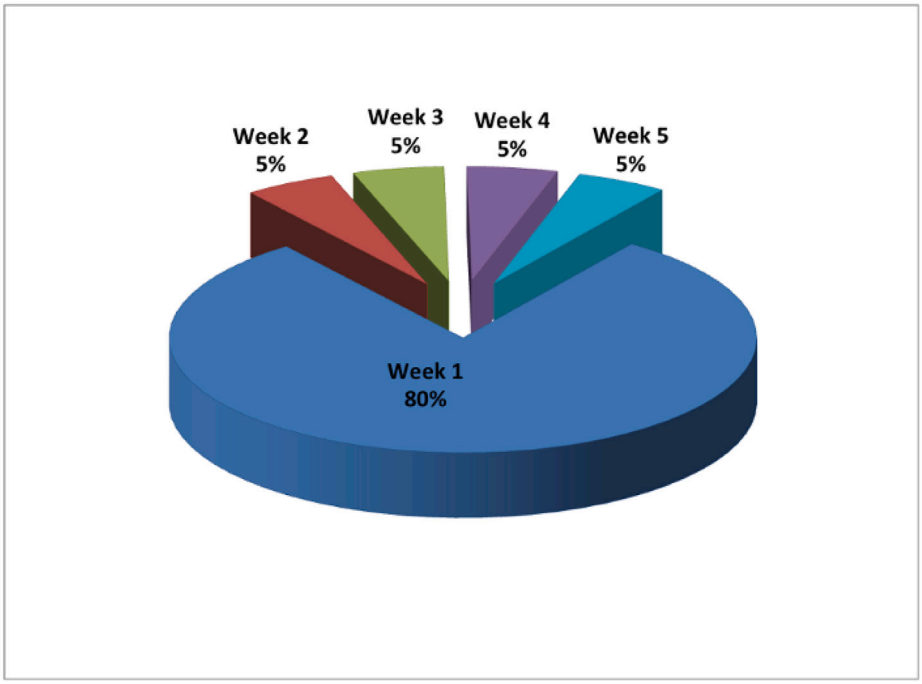
Your task is **to guess how much these participants improve in the test** as a result of studying with Duolingo.

Since we are interested in how your guess changes as you study through Duolingo, we will ask you to tell us your guess every week.

According to which question will you get paid your bonus payment?

At the end of the study, **one of your guesses** will be chosen at random for payment and your bonus payment will be determined according to the accuracy of your answer in that question.

The chart below shows how likely it is that a question from each weekly survey will be drawn for payment.



That is, there is an 80% chance that one of your guesses in WEEK 1 will determine your payment. There is 5% chance that one of your guesses in WEEK 2 will determine your payment. And so on.

Fig. A.2. Instructions about determination of bonus payment.

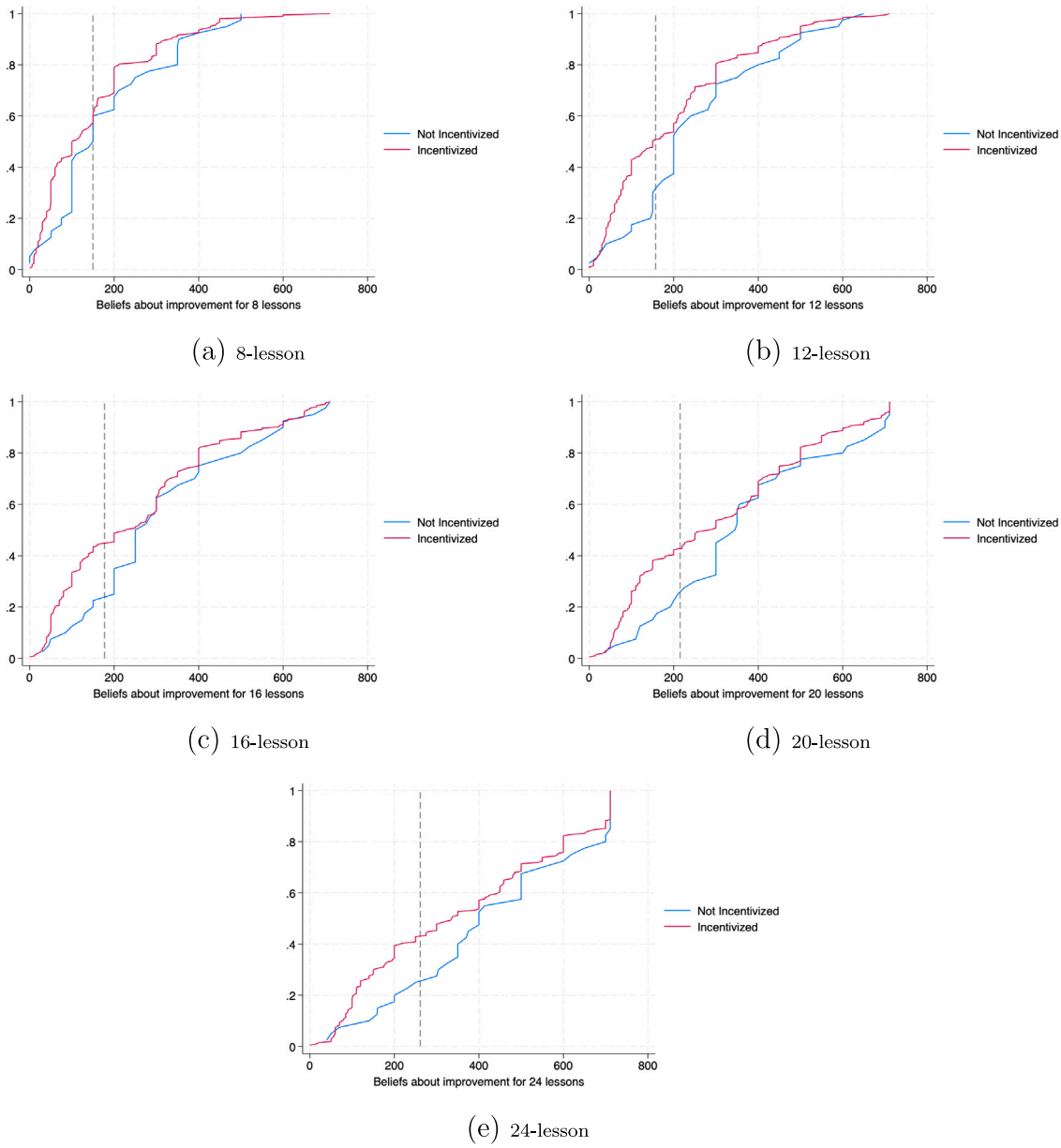


Fig. A.3. Distribution of baseline beliefs with and without incentivization. Notes: Gray dashed lines are the accuracy benchmarks estimated using data from Ersoy (2021).

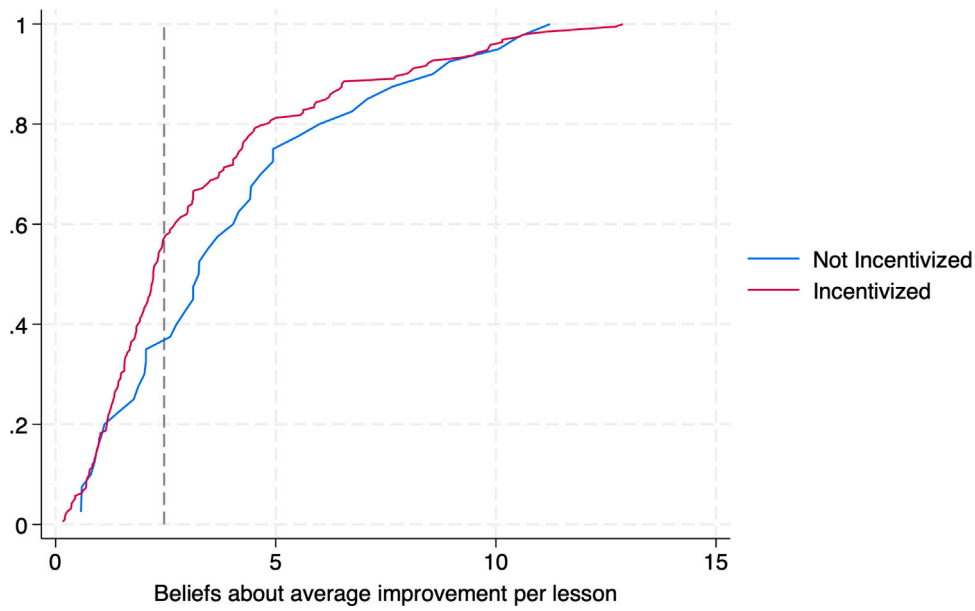


Fig. A.4. Distribution of week 1 beliefs with and without incentivization. Notes: Gray dashed line is the accuracy benchmark estimated using data from Ersoy (2021).

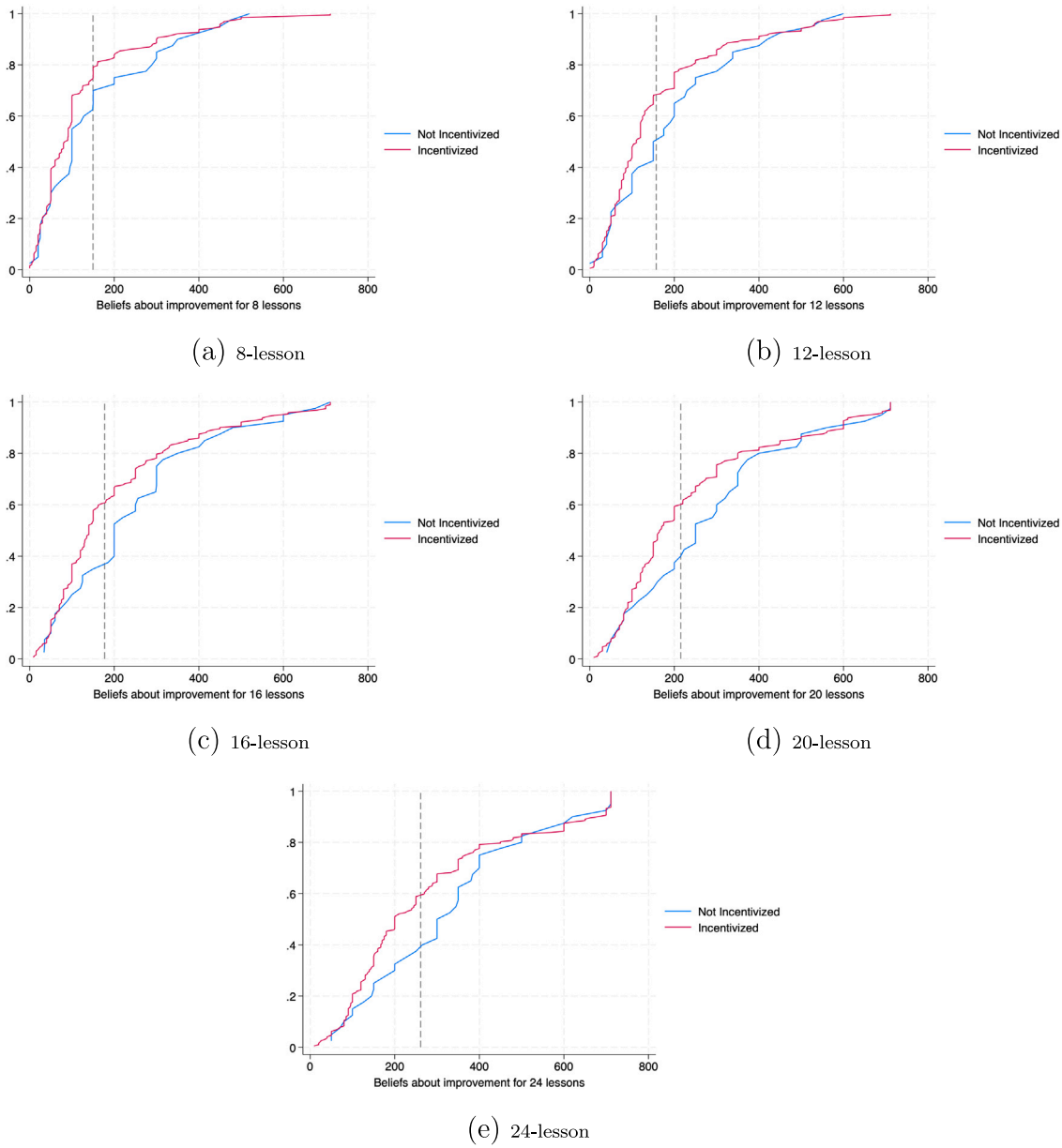


Fig. A.5. Distribution of Week 1 Beliefs with and without Incentivization. Notes: Gray dashed lines are the accuracy benchmarks estimated using data from Ersoy (2021).

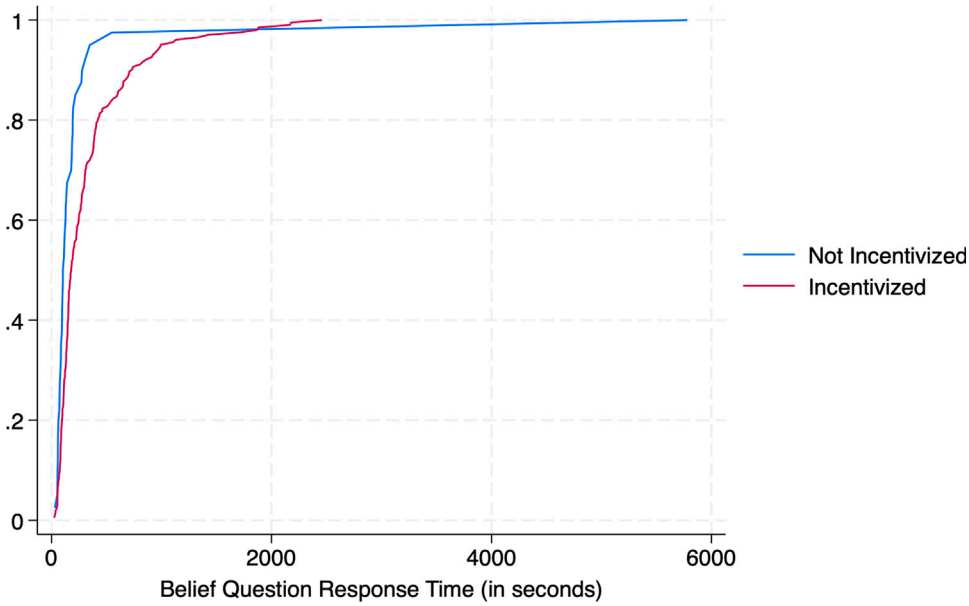


Fig. A.6. Distribution of response times at the baseline.

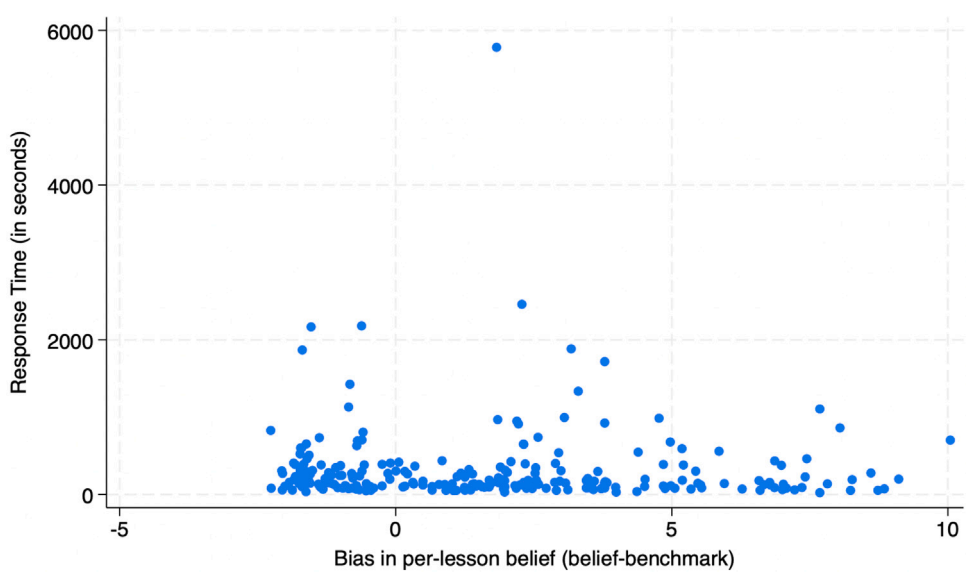


Fig. A.7. Correlation between response times and bias in beliefs at the baseline.

Table A.1
Summary statistics and balance.

	(1) All Sample	(2) Flat-Pay	(3) Incentives	(4) p-Value
Demographic Controls:				
Age	23.98	23.34	24.10	0.419
Female	0.76	0.80	0.75	0.427
African American	0.07	0.02	0.08	0.236
Asian	0.31	0.46	0.28	0.018
Caucasian	0.42	0.41	0.42	0.970
Hispanic	0.11	0.07	0.12	0.369
Other Ethnicity	0.09	0.02	0.11	0.095
Education Controls:				
GPA	3.38	3.24	3.40	0.064
ACT/SAT Math	0.09	0.07	0.10	0.723
ACT/SAT Reading	0.09	0.07	0.09	0.874
ACT/SAT Missing	0.50	0.54	0.50	0.650
Mother's Years of Education	14.33	14.24	14.35	0.878
Father's Years of Education	14.35	14.54	14.31	0.757
Freshman	0.07	0.05	0.08	0.491
Sophomore	0.14	0.12	0.14	0.749
Junior	0.23	0.22	0.23	0.833
Senior	0.32	0.44	0.30	0.083
Masters	0.17	0.12	0.17	0.416
PhD	0.05	0.05	0.05	0.940
Other Cohort	0.02	0.00	0.02	0.378
Humanities and Social Sciences	0.31	0.29	0.32	0.739
Business and Economics	0.09	0.15	0.08	0.139
Health and STEM	0.52	0.56	0.52	0.603
Other Major or Undecided	0.07	0.00	0.09	0.047
Initial Spanish Test Score	309.48	314.56	308.50	0.817
Personality Traits:				
Extroversion	6.29	6.34	6.28	0.844
Agreeableness	7.48	7.34	7.50	0.520
Neuroticism	6.04	6.17	6.01	0.611
Conscientiousness	7.42	7.00	7.50	0.043
Openness	7.46	7.44	7.47	0.909
Growth Mindset	29.09	29.37	29.03	0.573
Self-Control	43.33	40.68	43.84	0.020
Locus of Control	3.67	3.90	3.62	0.305
Risk Aversion	0.50	0.63	0.47	0.731
Time Inconsistency	1.04	1.05	1.03	0.670
Patience	1.00	1.00	1.00	0.889
Preference Survey Missing	0.08	0.05	0.09	0.391
Other Controls:				
Commitment	0.82	0.80	0.82	0.800
Interest	0.87	0.88	0.86	0.808
Spanish: Do not Know	0.39	0.46	0.38	0.320
Spanish: Beginner	0.53	0.46	0.54	0.371
Spanish: Elementary	0.07	0.07	0.07	0.950
Spanish: Intermediate	0.01	0.00	0.01	0.535
Native English Speaker	0.74	0.73	0.75	0.843
Serious Learner	0.51	0.54	0.51	0.730
Annual Family Income	81,535	103,232	77,359	0.086
Monthly Expenditure	277	263	280	0.759
Item Non-Response				
Baseline Belief Survey Missing	0.04	0.02	0.05	0.518
Week 1 Belief Survey Missing	0.09	0.02	0.10	0.123
Week 2 Belief Survey Missing	0.15	0.15	0.15	0.949
Week 3 Belief Survey Missing	0.17	0.15	0.17	0.722
Week 4 Belief Survey Missing	0.22	0.17	0.23	0.439

Sample consists of 254 students who completed the initial survey, who took the initial test, and who joined Duolingo. Column 1 presents the averages for all sample and columns 2 and 3 present the averages for the students who were randomly assigned to the flat-pay and incentives conditions, respectively. Column 4 presents the p-values from the two-sided t-test for equality of means across the two conditions. Extroversion, Agreeableness, Neuroticism, Conscientious and Openness are based on a short version of Big Five Personality Trait Survey (Rammstedt & John, 2007) and the score ranges between 2 and 10 for each of them. Growth Mindset is based on Growth Mindset Survey (Paunesku et al., 2015) and a higher score signifies growth mindset (range: 8 to 48). Self Control variable is based on Self Control Survey (Tangney et al., 2004) and ranges between 13 and 65. Locus of Control variable is based on Locus of Control Survey (Rotter, 1966) and ranges between 0 and 8, higher number indicating an external locus of control. Preference Survey controls are estimated using Convex Time Budget Sets (Andreoni & Sprenger, 2012). Time inconsistency parameter below 1 indicates present-biasedness. Commitment is equal to 1 if the student states to be highly committed to learn Spanish and Interest is equal to 1 if the student states to be highly interested in learning Spanish. Serious Learner is a dummy variable which is equal to 1 if a student is learning Spanish for studying, living, working abroad purposes or for improving employment opportunities and 0 otherwise.

Table A.2
Effects of incentives on accuracy of beliefs.

	(1) Baseline	(2) Week 1	(3) Week 2	(4) Week 3	(5) Week 4
Panel A: No Controls					
Incentives	−0.428	−0.437	−0.293	−0.220	0.0844
	(0.412)	(0.381)	(0.317)	(0.294)	(0.249)
Constant	2.993***	2.897***	2.665***	2.373***	2.113***
	(0.383)	(0.508)	(0.403)	(0.402)	(0.351)
Panel B: Demographic and Education Controls					
Incentives	−0.399	−0.200	−0.193	−0.0461	0.259
	(0.406)	(0.387)	(0.333)	(0.294)	(0.282)
Constant	1.409	4.974**	4.710***	2.815**	3.828***
	(2.148)	(1.979)	(1.265)	(1.325)	(1.404)
Panel C: All Controls					
Incentives	−0.198	−0.159	−0.188	0.0385	0.236
	(0.419)	(0.410)	(0.367)	(0.342)	(0.322)
Constant	−14.40	−10.96	15.27	3.132	18.01
	(17.67)	(15.71)	(13.89)	(14.55)	(14.21)
Observations	243	232	216	212	199

Dependent variable is the absolute difference between belief about improvement in test scores and the benchmark. Different columns refer to belief elicitation at different weeks. All panels & columns except column 1 include information treatment dummies. Panel A does not include any additional controls. Panel B includes demographic and education controls listed in Table 1. Panel C includes demographic controls, education controls, personality traits, and other controls listed in Table 1. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3
Effect of incentives on confidence in beliefs.

	(1) Baseline	(2) Week 1	(3) Week 2	(4) Week 3	(5) Week 4
Panel A: No Controls					
Incentives	0.00135	−0.123	−0.00739	−0.0437	−0.134
	(0.0859)	(0.0773)	(0.0848)	(0.0827)	(0.0839)
Constant	0.575***	0.630***	0.445***	0.537***	0.598***
	(0.0785)	(0.107)	(0.107)	(0.108)	(0.108)
Panel B: Demographic and Education Controls					
Incentives	−0.00531	−0.174**	0.0133	−0.0574	−0.162*
	(0.0866)	(0.0794)	(0.0885)	(0.0945)	(0.0893)
Constant	−0.705	0.158	−0.931**	−0.0417	0.393
	(0.444)	(0.451)	(0.470)	(0.474)	(0.555)
Panel C: All Controls					
Incentives	−0.0216	−0.204***	0.0226	−0.0899	−0.148
	(0.0890)	(0.0779)	(0.0958)	(0.101)	(0.0917)
Constant	0.0472	1.422	1.349	5.465	−0.160
	(3.945)	(4.121)	(4.367)	(3.692)	(4.628)
Observations	243	232	216	212	199

Dependent variable, confidence in beliefs in a given week, is equal to 1 if the student chose "extremely accurate", "very accurate", "moderately accurate" as answer to the confidence question and is equal to 0 if "slightly accurate" or "not accurate at all" was chosen by the student. Different columns refer to belief elicitation at different weeks. All panels & columns except column 1 include information treatment dummies. Panel A does not include any additional controls. Panel B includes demographic and education controls listed in Appendix Table A.1. Panel C includes demographic controls, education controls, personality traits, and other controls listed in Table 1. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

References

Andreoni, James, & Sprenger, Charles (2012). Estimating time preferences from convex budgets. *American Economic Review*, *102*(7), 3333–3356.

Armantier, Olivier, & Treich, Nicolas (2013). Eliciting beliefs: Proper scoring rules, incentives, stakes and hedging. *European Economic Review*, *62*, 17–40. http://dx.doi.org/10.1016/j.euroecorev.2013.03.008, URL https://www.sciencedirect.com/science/article/pii/S001429211300041X.

Blanco, Mariana, Engelmann, Dirk, Koch, Alexander K., & Normann, Hans-Theo (2010). Belief elicitation in experiments: is there a hedging problem? *Experimental Economics*, *13*(4), 412–438. http://dx.doi.org/10.1007/s10683-010-9249-1.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, *78*(1), 1–3.

Charness, Gary, Gneezy, Uri, & Rasocha, Vlastimil (2021). Experimental methods: Eliciting beliefs. *Journal of Economic Behavior and Organization*, *189*, 234–256. http://dx.doi.org/10.1016/j.jebo.2021.06.032, URL https://www.sciencedirect.com/science/article/pii/S0167268121002717.

Danz, David, Vesterlund, Lise, & Wilson, Alistair J. (2022). Belief elicitation and behavioral incentive compatibility. *American Economic Review*, *112*(9), 2851–2883.

Ersoy, Fulya (2021). Returns to effort: experimental evidence from an online language platform. *Experimental Economics*, *24*(3), 1047–1073. http://dx.doi.org/10.1007/s10683-020-09689-1.

Ersoy, Fulya (2023). Effects of perceived productivity on study effort: Evidence from a field experiment. *Journal of Economic Behavior and Organization*, *207*, 376–391. http://dx.doi.org/10.1016/j.jebo.2023.01.021, URL https://www.sciencedirect.com/science/article/pii/S0167268123000227.

Fuster, Andreas, Perez-Truglia, Ricardo, Wiederholt, Mirko, & Zafar, Basit (2022). Expectations with endogenous information acquisition: An experimental investigation. *The Review of Economics and Statistics*, *104*(5), 1059–1078. http://dx.doi.org/10.1162/rest_a_00994.

Gächter, Simon, & Renner, Elke (2010). The effects of (incentivized) belief elicitation in public goods experiments. *Experimental Economics*, *13*(3), 364–377. http://dx.doi.org/10.1007/s10683-010-9246-4.

Grewenig, Elisabeth, Lergetporer, Philipp, Werner, Katharina, & Woessmann, Ludger (2022). Incentives, search engines, and the elicitation of subjective beliefs: Evidence from representative online survey experiments. *Journal of Econometrics*, *231*(1), 304–326. http://dx.doi.org/10.1016/j.jeconom.2020.03.022, URL https://www.sciencedirect.com/science/article/pii/S0304407620302797. Annals Issue: Subjective Expectations & Probabilities in Economics.

Haaland, Ingar, Roth, Christopher, & Wohlfart, Johannes (2023). Designing information provision experiments. *Journal of Economic Literature*, *61*(1), 3–40. http://dx.doi.org/10.1257/jel.20211658, URL https://www.aeaweb.org/articles?id=10.1257/jel.20211658.

Hossain, Tanjim, & Okui, Ryo (2013). The binarized scoring rule. *Review of Economic Studies*, *80*(3 (284)), 984–1001, URL http://www.jstor.org/stable/43551453.

Jose, Victor Richmond R. (2016). Percentage and relative error measures in forecast evaluation. *Operations Research*, *65*(1), 200–211. http://dx.doi.org/10.1287/opre.2016.1550.

Manski, Charles F. (2004). Measuring expectations. *Econometrica*, *72*(5), 1329–1376.

Paunesku, David, Walton, Gregory M., Romero, Carissa, Smith, Eric N., Yeager, David S, & Dweck, Carol S. (2015). Mind-set interventions are a scalable treatment for academic underachievement. *Psychological Science*, *26*, 784–793.

Rammstedt, Beatrice, & John, Oliver P. (2007). Measuring personality in one minute or less: A 10-item short version of the big five inventory in english and german. *Journal of Research in Personality*, *41*(1), 203–212. http://dx.doi.org/10.1016/j.jrp.2006.02.001, URL http://www.sciencedirect.com/science/article/pii/S0092656606000195.

Roth, Christopher, & Wohlfart, Johannes (2020). How Do Expectations about the Macroeconomy Affect Personal Expectations and Behavior? *The Review of Economics and Statistics*, *102*(4), 731–748. http://dx.doi.org/10.1162/rest_a_00867, URL http://dx.doi.org/10.1162/rest_a_00867.

Rotter, J. B. (1966). Generalized expectancies for internal versus external control of reinforcement. *Psychological Monographs: General and Applied*, *80*, 1–28.

Schlag, Karl H., Tremewan, James, & van der Weele, Joël J. (2015). A penny for your thoughts: a survey of methods for eliciting beliefs. *Experimental Economics*, *18*(3), 457–490. http://dx.doi.org/10.1007/s10683-014-9416-x.

Schotter, Andrew, & Trevino, Isabel (2014). Belief elicitation in the laboratory. *Annual Review of Economics*, *6*, 103–128. http://dx.doi.org/10.1146/annurev-economics-080213-040927, URL https://www.annualreviews.org/content/journals/10.1146/annurev-economics-080213-040927.

Stantcheva, Stefanie (2021). Understanding Tax Policy: How do People Reason?*. *The Quarterly Journal of Economics*, *136*(4), 2309–2369. http://dx.doi.org/10.1093/qje/qjab033, URL http://dx.doi.org/10.1093/qje/qjab033.

Stantcheva, Stefanie (2023). How to run surveys: A guide to creating your own identifying variation and revealing the invisible. *Annual Review of Economics*, *15*(Volume 15, 2023), 205–234. http://dx.doi.org/10.1146/annurev-economics-091622-010157, URL https://www.annualreviews.org/content/journals/10.1146/annurev-economics-091622-010157.

Tangney, June P., Baumeister, Roy F., & Boone, Angie Luzio (2004). High self-control predicts good adjustment, less pathology, better grades, and interpersonal success. *Journal of Personality*, *72*(2), 271–324. http://dx.doi.org/10.1111/j.0022-3506.2004.00263.x, URL http://dx.doi.org/10.1111/j.0022-3506.2004.00263.x.

Wang, Stephanie W. (2011). Incentive effects: The case of belief elicitation from individuals in groups. *Economics Letters*, *111*(1), 30–33. http://dx.doi.org/10.1016/j.econlet.2010.11.045, URL https://www.sciencedirect.com/science/article/pii/S0165176510004180.

Zimmermann, Florian (2020). The dynamics of motivated beliefs. *American Economic Review*, *110*(2), 337–361. http://dx.doi.org/10.1257/aer.20180728, URL https://www.aeaweb.org/articles?id=10.1257/aer.20180728.